The University of Texas at Austin
In-Class Work 17

M339D Introduction to Financial Mathematics

November 06, 2026

Instructions: Provide your complete solution to the following problems. Final answers only, without appropriate justification, will receive zero points even if correct.

Log-Normal Stock Prices.

Problem 17.1: *In our usual notation, let* $\{S(t); t \geq 0\}$ *denote the stock price. The stock's mean rate of return is 10% while its volatility is* 0.30*. Then, ...*

a. $Var[\ln(S(t))] = 0.3t$
b. $Var[\ln(S(t))] = 0.09t^2$
c. $Var[\ln(S(t))] = 0.09t$
d. $Var[\ln(S(t))] = 0.09$
e. None of the above.

Solution: *The correct answer is* ***(c).***
The random variable $S(t)$ *is lognormal so that the random variable* $\ln(S(t))$ *is normal with variance* $0.3^2 t = 0.09t$*.*

Problem 17.2: *The current price of a non-dividend-paying stock is \$80 per share. Under the risk-neutral probability measure, its mean rate of return is 12% and its volatility is 30%.*

Let $R(0,t)$ *denote the realized return of this stock over the time period* $[0,t]$ *for any* $t > 0$*. Calculate* $\mathbb{E}^*[R(0,2)]$*.*

Solution:

$$(0.12 - 0.045)(2) = 0.15.$$

Problem 17.3: *A stock is valued at* $\$75.00$*. The continuously compounded, risk-free interest rate is* 10.0% *and the standard deviation of annualized returns is* 25.0%*. If the stock is lognormally distributed, what is the expected stock price after* 2 *years under the risk-neutral probability measure?*

Solution: *Let us denote the stock price today by* $S(0)$ *and that in three years by* $S(2)$*. According to the work we did in class, we need to calculate*

$$\mathbb{E}[S(2)] = S(0)e^{2r}$$

with r *equal to the continuously compounded risk-free interest rate. We are given in the problem that* $r = 0.10$*. So, the answer is* $75e^{0.20} \approx 91.605$*.*

Problem 17.4: *A non-dividend-paying stock is valued at \$55.00 per share. Its standard deviation of annualized returns is given to be 22.0%. The continuously compounded, risk-free interest rate is 12%. If the stock price is modeled using the lognormal distribution (as discussed in class), what is the median of the stock price in 3 years under the risk-neutral probability measure?*

Solution: *In our usual notation, it is given that* $S(0) = 55, r = 0.12$ *and* $\sigma = 0.22$. *As we have learned in class, under the risk-neutral probability measure, the median of the random variable* $S(3)$ *can be expressed as*

$$S(0)e^{(r-\frac{\sigma^2}{2})\times 3} = 73.31.$$

Problem 17.5: *Assume that the stock price is modeled using the lognormal distribution. The stock pays no dividends. Under* $\mathbb{P}^*$, *the annual mean rate of return on the stock is given to be* 12%. *Also under* $\mathbb{P}^*$, *the median time—*t *stock price is evaluated to be* $S(0)e^{0.1t}$. *What is the volatility parameter of this stock price?*

Solution:

$$r - \frac{\sigma^2}{2} = 0.12 - \frac{\sigma^2}{2} = 0.1 \quad \Rightarrow \quad \sigma = 0.2.$$

Problem 17.6: *The current stock price is* \$100 *per share. The stock price at any time* $t > 0$ *is modeled using the lognormal distribution. Assume that the continuously compounded risk-free interest rate equals 8%. Let its volatility equal 20%.*

Find the value t^* *at which the median stock price equals* \$120, *under the risk-neutral probability measure.*

Solution: *We know that the median stock price at time—*t *equals*

$$S(0)e^{(r-\frac{\sigma^2}{2})t}$$

under $\mathbb{P}^*$. *So,* t^* *must satisfy*

$$\ln\left(\frac{120}{S(0)}\right) = (r - \frac{\sigma^2}{2})t \quad \Rightarrow \quad t = \frac{\ln(120/100)}{0.12 - 0.04 - 0.02} = 3.0387.$$

Problem 17.7: *The volatility of the price of a non-dividend-paying stock is* 0.2. *The stock price is modeled using a log-normal distribution. Under* $\mathbb{P}^*$, *the expected time—*2 *stock price is \$120. What is the median of the time—*2 *stock price under* $\mathbb{P}^*$?

Solution: *The median of the time—*2 *stock price is*

$$\mathbb{E}^*[S(2)]e^{-\frac{2\sigma^2}{2}} = 120e^{-0.04} \approx 115.295.$$